

DEMYSTIFYING STORAGE VIRTUALISATION

Virtualisation of a data centre is incomplete without storage virtualisation, which has turned into a top priority for today's CIOs and data centre administrators. Fortunately, Linux has integrated storage virtualisation well with the kernel and user space.

Storage is the most expensive aspect of a data centre. Compared to server and networking technologies, standardisation in storage has lagged behind. This makes storage management an administrator's nightmare. Storage is also the slowest component and could lead to performance bottlenecks. Storage virtualisation is all about shielding where the data is stored, the way it is stored, and the type of storage.

The rationale and benefits of storage virtualisation are the same as that of server virtualisation. Server and desktop virtualisation have been the primary focus of data centre administrators and CIOs for more than a decade. Storage virtualisation has become mainstream only in the last few years.

In enterprises, information storage needs are increasing at the rate of 30 per cent annually. The lack of a clear picture regarding how resources are used results in data centres operating at only 30-40 per cent of their full capacity. The increase in storage capacity is not restricted to the cost of hardware purchased, but also additional operational costs for power, cooling and administration overheads. The principal financial motivation for storage virtualisation is to reduce costs without degrading performance and without additional complexity.

Virtualisation technologies like thin provisioning, data deduplication and automated tiering maximise storage utilisation and also enhance application throughput. Virtualisation also enhances the interoperability between vendor products. Though the storage virtualisation technologies have advanced significantly over the last few years, logical volume managers (LVM) have been around for more than a decade.

Different types of storage virtualisation

Virtualisation abstracts the physical storage devices and presents them as logical storage units for the application. The abstraction layer could be present in the host, the network or the storage device.

Host/server-based virtualisation through Logical Volume Managers (LVM): The Logical Volume Manager is a software layer between the file system and the operation system. This layer shields the complexity that lies below. Physically, the data might be stored on one or multiple disks based on the size and RAID technology used. Even a file of size 10 MB may span more than one disk. These operations might take a toll on the server's performance if all these operations are software based. Business critical applications that need higher performance methods such as RAID have hardware assistance from storage vendors.

To understand the LVM concept, we must first familiarise ourselves with the concepts of Physical Volume, Volume Group and Logical Volumes. Physical Volume (PV) is either the full hard disk or a portion of it. Volume Group (VG) is formed combining one or more PVs. Logical Volumes (LV) reside on the VGs for the file systems to exist. The interesting part of Logical Volumes is that we can create a file system that spans more than your largest hard disks in the pool.

LVM has been supported from version 2.4.x series of the Linux kernel. So, LVM is supported on most popular versions of Linux such as Ubuntu, RedHat and Linux Mint. There are two versions of LVM—LVM1 and LVM2. Between the two, LVM2 is more popular and has dependency on a module called